

E-Commerce Marketplace

Executive Summary Report

This report presents key findings from the analysis of the marketplace sales dataset, covering customer behaviour, product performance, geographic distribution, and shipping operations. It concludes with actionable recommendations to improve marketplace performance.

Key Performance Indicators

Metric	Value	Source
Total Sales	47.16bn	Executive Overview
Total Profit	5.49bn	Executive Overview
Total Customers	793	Customer Analysis
AOV	9.42M	Executive Overview
Total Orders	5,009	Shipping Analysis
Total Products	2K	Product Performance
Avg Delivery Days	4.08 days	Shipping Analysis
On Time Rate	78%	Shipping Analysis
Most Used Ship Mode	Standard Class	Shipping Analysis

1. Customer Analysis

Customer Metrics

Metric	Value	Notes
Active Customers	793	All customers active
New Customers	793	All customers new in period
Repeat Customers	781	98.5% made more than one order

Customers Over Time (Monthly)

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Customers	159	146	282	290	301	290	281	280	323	459	483	454

Key Findings

- 793 total customers — 781 (98.5%) made more than one order, indicating strong repeat behaviour.
- Customer volume peaks in October-November, suggesting seasonal buying patterns.
- Lowest activity recorded in January-February — opportunity for early-year campaigns.
- All three categories (Furniture, Office Supplies, Technology) show equal segment splits: 409 / 236 / 148.

RFM Segmentation

Segment	Description	Recommended Action
Champions	Recent, frequent, high spenders	Reward & retain
Loyal Customers	Regular buyers with solid spend	Upsell & cross-sell
Potential Loyalists	Recent, moderate frequency	Nurture with offers
At Risk	Past buyers going quiet	Win-back campaigns
Lost Customers	No purchases in 180+ days	Re-engagement emails

2. Product Performance

Category Performance

Category	Total Sales	Notes
Furniture	19.9bn	Highest sales — low margin risk
Technology	17.1bn	Strong balanced performance
Office Supplies	10.2bn	Steady, lower volume

Top 10 Products by Sales

Product	Sales
Riverside Palais Royal Lawyers...	3.1bn
Hon 4070 Series Pagoda Ar...	2.1bn
Global Troy Executive Leat...	1.9bn
Global Commerce Series Hi...	1.6bn
DMI Eclipse Executive Suite...	1.5bn
Sauder Forest Hills Library...	1.5bn
Apple iPhone 5	1.0bn

Top Sub-Categories by Sales

Sub-Category	Sales	Notes
Chairs	9.26bn	Largest sub-category
Phones	6.74bn	Top technology sub-category
Bookcases	6.06bn	Strong furniture performer
Machines	4.83bn	High-value items
Accessories	3.36bn	Volume-driven

Key Findings

- Furniture leads in sales (19.9bn) but profit margin needs monitoring — Tables sub-category shows negative profit.
- Technology maintains a healthy profit balance relative to sales volume.
- Scatter plot shows several sub-categories with negative profit — immediate pricing review needed.
- Apple iPhone 5 is the only single consumer product in the top 10.

3. Geographic Performance

Regional Sales Performance

Region	Total Sales	% of Total	Notes
East	28.6bn	61%	Dominant region — top performer
West	11.8bn	25%	Strong second region
Central	3.9bn	8%	Growth opportunity
South	2.8bn	6%	Weakest region — needs attention

Top Geographic Performers

Level	Winner	Metric
Region	East	28.6bn total sales
State	New York	4.6bn total profit
City	New York City	Top city by sales

Sales by Region and Segment

Region	Consumer	Corporate	Home Office
East	63.28%	26.59%	10.13%
West	61.77%	24.48%	13.75%
Central	65.63%	16.89%	17.47%
South	32.39%	22.29%	45.32%

Key Findings

- East region dominates with 61% of total sales — New York is the top state.
- South region is significantly underperforming at only 6% of total sales.
- South has an unusually high Home Office segment (45.32%) vs other regions.
- Consumer segment leads in all regions except South.

4. Shipping Analysis

Ship Mode Distribution

Ship Mode	Orders	% Share	Avg Delivery	Total Sales
Standard Class	3K	59.77%	5.0 days	32.4bn
Second Class	1K	19.25%	3.5 days	5.3bn
First Class	1K	15.71%	2.3 days	6.0bn
Same Day	0K	5.27%	0.0 days	1.0bn

Key Findings

- Standard Class dominates at 59.77% of all orders despite being the slowest (5 days).
- Average delivery time is 4.08 days across all ship modes.
- On Time Rate is 78% — meaning 22% of orders arrive late.
- Standard Class generates the most revenue (32.4bn) but has the longest delivery time.
- First Class generates more revenue (6.0bn) than Second Class (5.3bn) despite fewer orders.
- Same Day delivery is underutilised at only 5.27% of orders.

5. Business Recommendations

1. Improve On-Time Delivery Rate

With only 78% on-time rate, 22% of orders are arriving late. Review Standard Class logistics partners and set SLA targets. Offer First Class at discounted rates for high-value orders above a threshold.

2. Grow South Region

South region accounts for only 6% of total sales (2.8bn) — the weakest performer. The unusually high Home Office segment (45.32%) suggests an untapped corporate opportunity. Launch B2B targeted campaigns to grow the Corporate segment in the South.

3. Address Furniture Profitability

Furniture leads in sales (19.9bn) but shows dangerously low profit margins. Tables sub-category is loss-making based on scatter plot analysis. Cap discounts at 20% for Furniture and review Tables pricing immediately.

4. Promote Technology Category

Technology (17.1bn) has a healthier profit profile than Furniture. Increase marketing investment in Phones (6.74bn) and Machines (4.83bn). Bundle accessories with device purchases to increase AOV above the current 9.42M.

5. Leverage Peak Season

Customer activity peaks in October-November. Prepare inventory and logistics capacity 6-8 weeks ahead of Q4. Run early-access promotions in September to smooth demand spikes.

6. Re-engage Low-Season Customers

January-February show the lowest customer activity (159 and 146 respectively). Deploy targeted email campaigns with seasonal offers in January to maintain engagement. Focus win-back efforts on customers who last purchased in Q4 of the prior year.

This report was generated as part of the E-Commerce Marketplace Technical Assessment. All figures sourced directly from Power BI dashboard analysis.